

Stage-1 Range-FFT Engine

Design, Optimisation and Verification Report

Streaming FFT block for an FMCW / MIMO radar accelerator
Xilinx Zynq-7000 — xc7z020clg400-1 — Digilent Zybo Z7-20

Status	RTL complete, synthesised, implemented (routed), simulation verified
Throughput	100 MSPS per lane, 1 sample/clock, sustained
Resources	24 DSP48E1, 18 RAMB18, 3 909 LUT, 6 816 FF per lane
Timing	WNS +2.735 ns at 100 MHz (Fmax ~138 MHz), WHS +0.013 ns, met
Verification	7 frames, 28 checks, 0 errors
Not done	Never run on physical hardware

1 Purpose and context

This block is the **first processing stage** of an FMCW / MIMO radar signal chain. Its job is to convert a stream of complex baseband samples into a frequency spectrum, one transform per chirp, continuously and without dropping samples.

1.1 Why an FFT gives range

An FMCW radar transmits a tone whose frequency ramps linearly with time — a *chirp*. The echo from a target returns after a round-trip delay. Because the transmitter has moved on to a higher frequency in the meantime, mixing the echo against the transmitted signal produces a low-frequency *beat* whose frequency is proportional to target distance:

$$f_{\text{beat}} = (2 * R * S) / c$$

R = target range (m)
S = chirp slope (Hz/s)
c = 3e8 m/s

A Fourier transform of the beat signal therefore sorts targets by distance: **bin index maps directly to range**. This is called the *Range FFT*, and it is what this block computes.

1.2 Position in the full chain

```

ADC -> input buffer -> [ RANGE FFT ] -> corner turn -> Doppler FFT
                        ^                               -> AoA / spatial FFT
                        |                               -> detection (CFAR)
                        |
                    this block

```

The stages downstream extract velocity (Doppler, a transform across chirps) and angle (AoA, a transform across the virtual antenna array). **Both are coherent** — they recover target information entirely from the *phase* of each range bin. That fact drives the single most important interface decision in this design, covered in section 5.1.

1.3 Scope

The wider accelerator is divided into blocks owned by different people. This report covers **the FFT engine only**: the transform itself, its configuration and status handling, and its AXI-Stream boundaries.

In scope	Out of scope (other blocks)
Transform core instantiation and configuration	Window function (input stage)
Runtime transform-size switching	Magnitude / magnitude-squared
Block-floating-point exponent forwarding	Frame header and 256-bit packing
Status-channel handling	Width conversion, AXI DMA, DDR
Backpressure and boundary registering	Corner turn, Doppler, AoA
Lane replication (NUM_LANES)	Detection / CFAR

2 Specification

Every value below was fixed before design and traced to the project specification. None was chosen for implementation convenience.

Parameter	Value	Why this value
Device	xc7z020clg400-1	220 DSP / 140 BRAM36 / 53 200 LUT; Z7-10 too tight
Clock	100 MHz (FCLK_CLK0)	PS-derived, so no extra MMCM and no CDC to AXI-HP
Transform size	1024 and 2048, runtime	Range resolution vs frame rate trade, selectable live
FFT core	Xilinx FFT v9.1	Verified IP; custom RTL saves nothing (section 8)
Architecture	Pipelined Streaming I/O	Only xfft mode sustaining 1 sample/clock
Radix	Radix-2 (internally 2 ^{<sup>25</sup>}	Set by the architecture choice, not free
Input	complex {Q,I}, 16-bit Q1.15	IQ baseband; real-only would fold negative freqs
Twiddle	16-bit Q1.15	Matches data width; sets phase-noise floor
Scaling	Block Floating Point	Preserves dynamic range on weak returns
Rounding	Convergent (half-to-even)	Truncation biases toward $-\infty$, builds a DC term
Output order	Natural	Consumers index bins directly
Throughput	1 sample/clock = 100 MSPS Streaming, no gaps between frames	
Backpressure	Full, Non-Realtime throttle	Core stalls rather than dropping samples
Lanes	NUM_LANES parameter	One lane per RX antenna

2.1 Fixed-point format

Q1.15 means a 16-bit signed integer v represents the fraction $v / 32768$, giving a range of -1.0 to $+0.99997$. A sample at 0.4 full scale is stored as $\text{round}(0.4 \times 32767) = 13107$.

Block Floating Point keeps the output at the same 16-bit width and reports a shared exponent, BLK_EXP, for the whole frame. The true value of a bin is its 16-bit value scaled by $2^{\text{BLK_EXP}}$. This gives per-frame automatic gain: a weak chirp is scaled up rather than lost in the low bits.

Note for consumers: BLK_EXP applies to the *complex* output. If a downstream block squares the data to get power, the scale becomes $2^{2 \times \text{BLK_EXP}}$. Missing this gives a factor-of-two error in dB.

3 Architecture

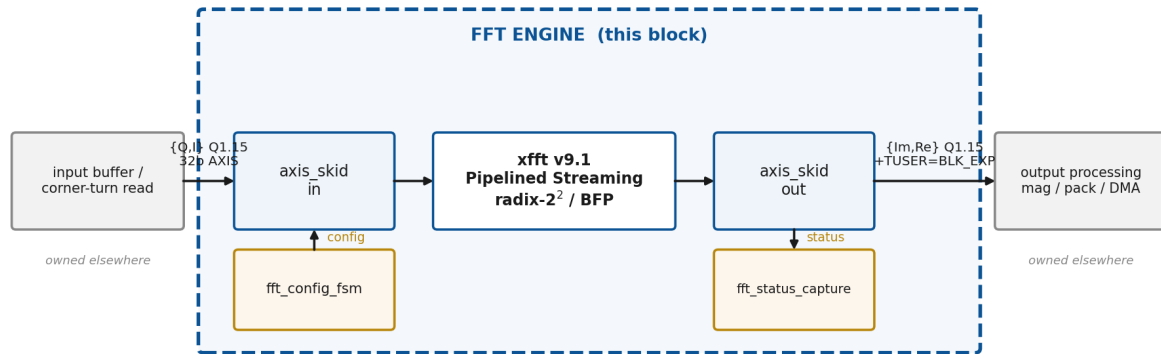


Figure 1 — FFT engine block diagram. Grey blocks are owned by other people.

The datapath is deliberately thin: two register slices around a vendor transform core. All the engineering is in configuration, status handling and boundary discipline, because that is where this class of design actually fails.

3.1 Module hierarchy

fft_engine_top	NUM_LANES replication, flattened buses
+ fft_lane [generate]	one lane per RX channel
+ axis_skid (in)	register slice, breaks treddy path
+ fft_config_fsm	config channel, runtime N, frame fencing
+ xfft_0	Xilinx FFT IP v9.1
+ fft_status_capture	status drain, owns m_axis_status_tready
+ axis_skid (out)	register slice + TUSER carry

3.2 Module reference

Module	Function	Latency	Cost
fft_engine_top	Structural. Replicates lanes, flattens per-lane streams into single vectors so the module stays instantiable from a block design.	0 cycles	200000
fft_lane	One complete lane. Owns the frame-active tracker that fences config writes to frame boundaries.	1 cycle	200000
axis_skid	AXI-Stream register slice. Full throughput (1 beat/clock), one cycle latency. Acts as a beat into a skid slot when the output is full.	1 cycle	200000
fft_config_fsm	Builds and issues the config word. Re-arms on transform-size change, issues only between frames, holds the sample stream.	1 cycle	200000
xfft_0	The transform. Pipelined streaming, radix-2 ² internally, BFP, 200000 RAMB16, natural order.	200000 cycles	200000
fft_status_capture	Drains the status channel unconditionally and latches BLK_EXP for CSR readback.	1 cycle	200000

3.3 Interface contract

This is the part that determines whether the block integrates cleanly. Everything else is internal.

Signal	Width	Meaning
s_axis_tdata	32	{Q[15:0], I[15:0]} Q1.15, imaginary in the upper half
s_axis_tvalid / treddy	1	Full backpressure; treddy must not be tied high
s_axis_tlast	1	Asserted on sample N-1 of a frame
m_axis_tdata	32	{Im[15:0], Re[15:0]} Q1.15, scaled by 2 ^{BLK_EXP}
m_axis_tuser	8	BLK_EXP, constant across all N beats of the frame
m_axis_tlast	1	Asserted on bin N-1
nfft_sel_i	1	0 = 1024, 1 = 2048. Latched at a frame boundary only
nfft_applied_o	5	log2(N) currently loaded in the core: 10 or 11
busy_o / blk_exp_dbg_o	1 / 8	Status and CSR readback; not the datapath

4 Throughput and latency

These are different quantities and are routinely confused. This design has **large latency and full throughput at the same time**.

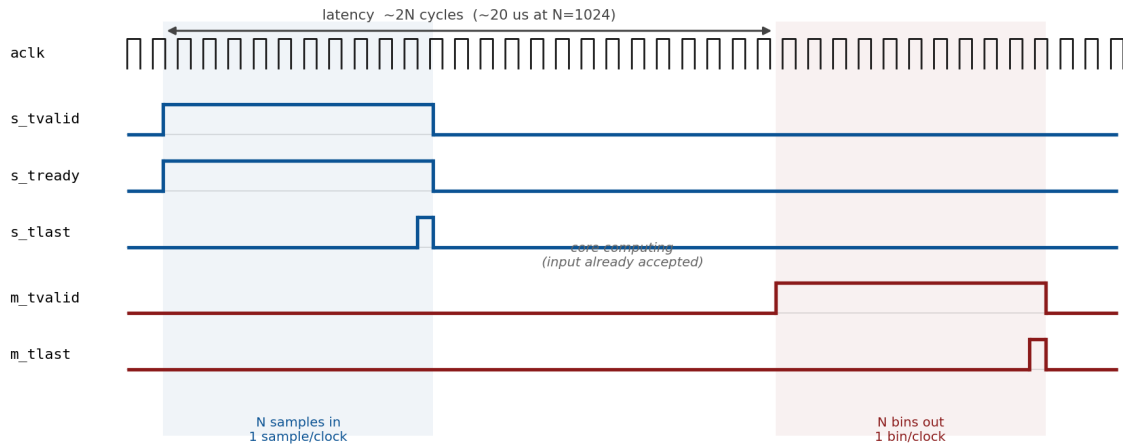


Figure 2 — AXI-Stream behaviour. The core accepts one sample every clock while producing nothing for roughly 2N cycles; output then streams at one bin per clock.

4.1 Throughput arithmetic

Quantity	N = 1024	N = 2048
Input sample rate	100.000 MSPS	100.000 MSPS
Input byte rate (4 B/sample)	400.000 MB/s	400.000 MB/s
Cycles to ingest one frame	1 024	2 048
Frame ingest time	10.24 us	20.48 us
Transforms per second	97 656.25	48 828.125
Output bin rate	100.000 Mbin/s	100.000 Mbin/s
Output byte rate (4 B/bin)	400.000 MB/s	400.000 MB/s
With 32-byte frame header	403.125 MB/s	401.563 MB/s

```
frames/s = 100e6 / 1024 = 97 656.25
bytes/s  = 97 656.25 x 4128 B = 403 125 000 = 403.125 MB/s
```

```
frames/s = 100e6 / 2048 = 48 828.125
bytes/s  = 48 828.125 x 8224 B = 401 562 500 = 401.563 MB/s
```

4.2 Latency

A pipelined streaming FFT holds roughly N–1 words in its stage delay lines and a further N in the natural-order reorder buffer:

```
L ~= (N - 1) stage delay lines
    + N      output reorder buffer
    + pipeline registers
    ~= 2N cycles
```

N=1024 -> ~2 048 cycles ~= 20.5 us

N=2048 -> ~4 096 cycles ~= 41.0 us

Latency does not reduce throughput. Frames pipeline back to back; the 20 us never accumulates. Latency matters only for closed-loop response, which this stage does not have. Optimising it would buy nothing.

5 Design decisions

5.1 The engine outputs complex bins, not magnitude

The original specification had this stage emit 32-bit magnitude-squared. That is incompatible with the rest of the stated chain. Doppler and AoA are coherent transforms operating on the phase of each range bin; $|X|^2$ is real and non-negative, so phase is destroyed and cannot be recovered. A magnitude output can only ever produce a single-chirp range profile.

The block therefore ends at the transform output and emits complex data. The cost of this is zero and the saving is real:

	magnitude-squared	complex passthrough
Bits per bin	32	32
Bytes per frame (N=1024)	4 128	4 128
DDR rate	403.125 MB/s	403.125 MB/s
DSP cost	2	0
Usable by Doppler / AoA	no	yes

Whether to square is now the downstream block's decision, which is where it belongs.

5.2 BLK_EXP travels on TUSER, not a side channel

The generated core reports `C_M_AXIS_DATA_TUSER_WIDTH = 8` with both optional output fields disabled (`C_HAS_XK_INDEX = 0`, `C_HAS_OVFLO = 0`). The only field that can occupy those bits is `BLK_EXP`, present because scaling is block floating point. The core therefore already presents the exponent beat-aligned with its own data.

An earlier revision of this design paired `BLK_EXP` to frames by hand using a small FIFO, on the assumption that the status beat arrives no later than the first data beat. That assumption was never verified. Taking the exponent from TUSER removes it entirely: alignment is guaranteed by the IP, and a downstream block physically cannot pair an exponent with the wrong frame.

5.3 Both boundaries are registered

A block that writes `assign s_tready = m_tready;` forwards the downstream ready combinationaly. That is legal AXI-Stream and harmless in one module. Chain five separately-owned blocks that each do it and `tready` becomes one combinational path from DDR back to the ADC — and no single owner can debug it, because the path belongs to everyone.

Each lane is bracketed by skid buffers. Cost is two cycles of latency and roughly 140 flip-flops. The return is that this block's timing is self-contained and provable in isolation, which is what the +2.735 ns result in section 9 actually measures.

5.4 Runtime size switching is fenced to frame boundaries

The configuration FSM re-arms whenever `nfft_sel` changes, but issues the new config word only when no frame is in flight, and holds the sample stream off until the core accepts it. A config write mid-frame corrupts the transform in progress.

6 Where the throughput limit actually is

A common mistake is to optimise the FFT because it is the visible compute block. Rate-matching every stage shows the FFT has no recoverable slack, and that the constraint sits elsewhere.

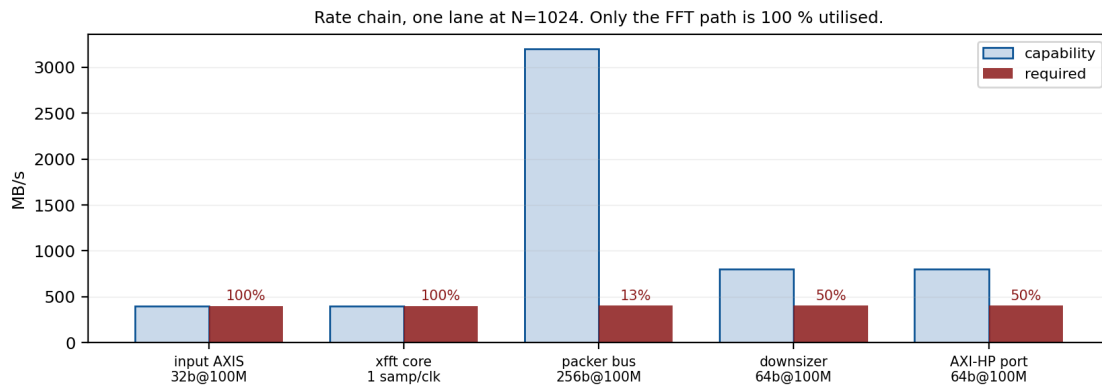


Figure 3 — Rate chain for one lane at N=1024.

Stage	Capability	Required	Utilisation
Input AXIS, 32 b @ 100 MHz	400 MB/s	400 MB/s	100 %
xfft core, 1 sample/clock	100 MSPS	100 MSPS	100 %
Packer bus, 256 b @ 100 MHz	3 200 MB/s	403 MB/s	12.6 %
Downsizer / 64 b AXIS	800 MB/s	403 MB/s	50.4 %
AXI-HP (AFI) port, 64 b @ 100 MHz	800 MB/s	403 MB/s	50.4 %

The FFT and the input path are 100 % utilised **by construction** — a 1 sample/clock pipe fed at 1 sample/clock is exactly balanced, not congested. There is no throughput being lost anywhere.

6.1 The wall appears at the second lane

```

2 lanes x 403.125 MB/s           = 806.25 MB/s
one 64-bit AXI-HP port @ 100 MHz = 800.00 MB/s
-----
overflow by                       0.78 %

```

Two lanes miss a single AXI-HP port by under one percent — and that is the theoretical port figure, before AXI address phases and DMA descriptor fetches. In practice one port carries one lane. Zynq-7020 has four AXI-HP ports, so the system ceiling is approximately **four lanes, 400 MSPS aggregate**, and that consumes the entire PL-to-PS interface.

Conclusion. The FFT engine is not the bottleneck and is not close to being one. In order, the real limits are: Stage-2 corner-turn DDR bandwidth, then AXI-HP port bandwidth (~4 lanes), then the DSP budget (9 lanes), and only then the transform itself.

7 Optimisation analysis

Each option below was evaluated against measured or derived numbers, not intuition. Several were rejected, and one earlier claim was withdrawn after measurement.

7.1 The IP is already radix-2²

The most significant finding. Synthesis instantiates six complex multipliers (24 DSP / 4 each) and the elaboration log contains r22_memory, r22_pe and r22_bf. Xilinx's "Radix-2, Pipelined Streaming" is internally **radix-2 squared**.

```
plain radix-2 SDF, N=2048 : log2(2048) - 1      = 9 non-trivial multipliers
radix-2^2 SDF,   N=2048 : ceil(log4(2048))    = 6 non-trivial multipliers
measured        : 24 DSP / 4 per cmpy = 6
```

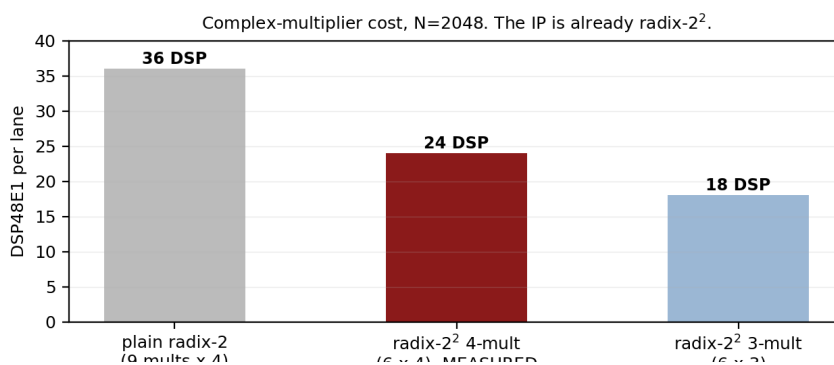


Figure 4 — Complex-multiplier cost per lane, N=2048.

The radix-2² decomposition makes every second twiddle a multiplication by $-j$, which reduces to swapping real and imaginary parts and negating — free in hardware. **A hand-written radix-2² SDF would therefore save no DSPs over this IP**, at identical 1 sample/clock throughput, while requiring a from-scratch block-floating-point implementation. An earlier version of this analysis claimed a ~44 % DSP saving for custom RTL; that claim is withdrawn.

7.2 Options evaluated

Option	Effect	Verdict
3-multiplier complex structure	24 − 18 DSP per lane (−25%); changes to 4 DSPs; allows 9 lanes; bandwidth allows 4. Buys DSPs that can be used for other things.	Adopted.
Delay lines in Block RAM (5 stages) moves ~1 984 of 2 047 words from SDF to BRAM	Adopted. A 1024-word x 32-bit delay is ~1 RAMB36 versus ~1 024 LUTs.	Adopted.
Registered block boundaries	+2 cycles latency, ~140 FF	Adopted. Timing isolation across an integration boundary.
Bit-reversed output	Saves ~4 BRAM36 and ~N cycles of latency	Rejected. Violates the frozen natural-order contract.
150 MHz DMA clock domain	2 lanes per AXI-HP port instead of 1	Rejected. Introduces a clock domain not in the specification.
16-bit log-magnitude output	Halves output bandwidth	Rejected. Destroys phase; belongs after Doppler, not here.
Custom radix-2 ² SDF	8 DSP saving, same throughput	Rejected. ‐6 months and a BFP reimplementaion for no gain.
Custom multi-path MDC	> 1 sample/clock on one stream	Rejected. MIMO gives per-antenna parallelism for free; P independent of N.

7.3 How optimised is this, honestly

At the block level the design is at the practical optimum for its specification:

- Throughput is 1 sample/clock, which is the maximum the chosen architecture supports and the maximum a single AXI-HP port can carry away.
- DSP cost is 24 per lane, matching the radix-2² lower bound for six non-trivial twiddle stages at 4 multipliers each.
- Timing closes with 38 % margin, so there is no pressure to trade area for speed.

- The block occupies ~11 % of the device's DSPs and ~7 % of its logic, leaving room for the eight other blocks in the accelerator.

The honest caveat: further optimisation of this block would be wasted effort. It is not the system bottleneck, and roughly half the theoretical lane count the silicon permits is already unreachable because of interface bandwidth.

8 Verification

The testbench is self-checking and has no external dependencies — stimulus is generated with SystemVerilog real math, so there are no data files to keep in sync and no Python required to reproduce the result.

8.1 Stimulus

A complex tone at bin k : $x[n] = A \cdot \exp(j2\pi kn/N)$, with $A = 0.4$ full scale. The amplitude is deliberately 0.4 and not 0.5: at 0.5 a full-scale tone sits exactly on the block-floating-point overflow boundary, where the ideal peak lands one LSB above Q1.15 full scale and *correct* hardware fails the check.

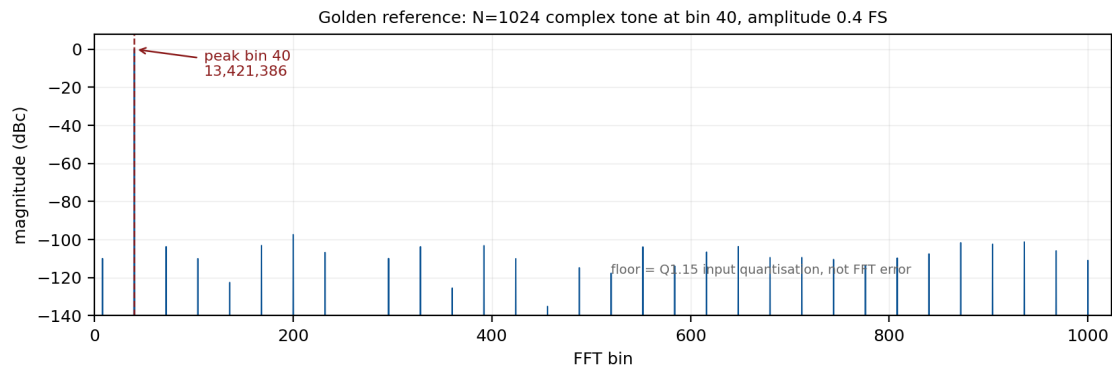


Figure 5 — Golden reference spectrum. The noise floor is Q1.15 input quantisation, not transform error.

8.2 Checks performed per frame

ID	Check	Catches
C1	Beat count equals N	Wrong transform size, dropped or duplicated beats
C2	TLAST asserted once, on the final beat	Frame boundary errors
C3	TUSER constant across all N beats	BLK_EXP mis-pairing or mid-frame change
C4	No X or Z on any output beat	Uninitialised logic, reset problems
C5	Peak bin equals the injected tone bin	Config word layout, transform direction, size

C5 is the load-bearing check. The configuration word field layout could not be confirmed from the generated files — only its 16-bit width could. A wrong NFFT field gives the wrong transform size and moves the peak; a wrong FWD_INV field gives an inverse transform and mirrors the peak to $N-k$. Either failure is caught here, and nowhere in synthesis.

8.3 Results

Case	N	Gaps	Stalls	Tone bin	Peak bin	BLK_EXP	Beats	Result
Clean	1024	0 %	0 %	40	40	9	1024	pass
Source gaps	1024	30 %	0 %	137	137	9	1024	pass
Sink stalls	1024	0 %	40 %	300	300	9	1024	pass
Both	1024	25 %	50 %	511	511	9	1024	pass
After size switch	2048	0 %	0 %	80	80	10	2048	pass
Both	2048	25 %	50 %	777	777	10	2048	pass
Switched back	1024	0 %	0 %	40	40	9	1024	pass

frames: 7 checks: 28 errors: 0 RESULT: PASS

8.4 BLK_EXP verified independently

The exponent was not merely checked for constancy — it was verified numerically. Block floating point selects the smallest shift that avoids 16-bit overflow:

```
input amplitude = 0.4 x 32767 = 13 107
```

```
N=1024: 13107 x 1024 / 2^9 = 26 214 (2^8 would give 52 428 -> overflow)
```

```
N=2048: 13107 x 2048 / 2^10 = 26 214 (2^9 would give 52 428 -> overflow)
```

BLK_EXP came out as 9 and 10 respectively — in both cases the minimum non-overflowing shift — and both give the same peak magnitude, exactly 0.4×2^{16} . This confirms the BFP path and the TUSER plumbing together.

8.5 What is not verified

Stated plainly, because overclaiming verification is worse than having less of it. The coverage is **structural checks plus peak-bin position on a single complex tone**. There is no golden-model spectral comparison bin by bin, and no impulse, DC, two-tone, off-bin leakage, broadband noise or low-amplitude cases. This is not a two-tier regression and should not be described as one.

9 Implementation results

Vivado 2025.1, xc7z020clg400-1, out-of-context, NUM_LANES = 1, routed.

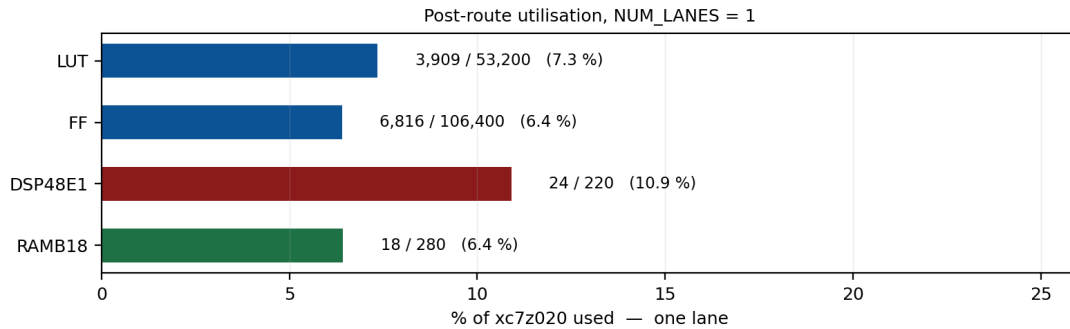


Figure 6 — Post-route utilisation for one lane.

Resource	Post-route	Synthesis	Available	% used	Lanes this allows
LUT	3 909	4 036	53 200	7.3 %	13
FF	6 816	6 871	106 400	6.4 %	15
DSP48E1	24	24	220	10.9 %	9
RAMB18	18	18	280	6.4 %	15
RAMB36	0	0	140	0 %	—

Wrapper logic — two skid buffers, the config FSM and the status drain — accounts for only about 163 flip-flops and 90 LUTs of that total. The remainder is the transform core.

9.1 Timing

WNS = +2.735 ns setup, post-route -> critical path 7.265 ns, Fmax ~138 MHz
WHS = +0.013 ns hold, post-route -> MET

38 % setup margin at the specified 100 MHz. The synthesis estimate was +5.824 ns; the drop is routing delay becoming real, which is expected and not a concern.

9.2 On the hold figure

+0.013 ns looks alarming and is not. The worst hold paths are internal FDRE → RAMB18E1 data-input paths inside the transform core, with zero logic levels. Two things make this the normal result:

- The router fixes hold by adding routing detour and stops the moment slack goes positive, because hold padding costs setup margin. Small positive WHS is the signature of a route that succeeded, not one that barely survived.
- BRAM data pins have large hold requirements — 0.296 ns here, against roughly 0.03 ns for a slice flip-flop. Flip-flop-to-BRAM paths are the tightest hold paths in almost any design using block RAM.

It is also reported at the Fast Process Corner, which is already the pessimistic corner for hold.

Caveat that does survive: HD.CLK_SRC is unset in out-of-context mode, so clock skew (0.022 ns here) is computed on an estimated clock net rather than a real global buffer tree. Because larger destination-clock delay directly consumes hold slack, WHS must be re-confirmed after integration with the PS clock on a real BUFG. If it goes negative there the fix is clocking and placement constraints, not RTL — the paths are inside the IP.

10 Defects found and corrected

Found by review and simulation of this block, in the order they surfaced.

#	Defect	Consequence	Resolution
1	Stage-1 output specified as magnitude-squared; Doppler and AoA stages took series complements; squaring moved downstream	Distorts phase	Block series complements
2	Config FSM latched itself off after the first size check after reset; runtime 1024/2048 switching impossible, fenced to frame boundaries	Frame size stuck	Reset to 1024
3	Status channel tready left undriven	Status FIFO fills; core trips an internal assertion; captures around the frame and holds it high	Set status to 1 before frame
4	BLK_EXP paired to frames by hand	Relied on an unverified assumption about taking from axis_data_tuser, aligned by the core	Take from axis_data_tuser
5	Combinational tready straight through	Could be holding tready path across the whole acceleration	Both accelerators registered with skid buffers
6	transform_length set to 1024 with 2048 time bins	Physically unbuildable; would be 1024 (the minimum), 1024 selected at runtime	Set to 2048 (the maximum)

Defect 6 is worth dwelling on. Vivado validated the property and accepted the value — it was legal, and wrong. Tooling catches malformed input; it does not catch well-formed input that means the wrong thing.

10.1 Corrections made to this analysis

Recorded because an analysis that never revises itself is not being checked:

Earlier claim	Correction	Source
Custom radix-2 ²	50% draw five H2s and 50% draw 2 super H2s	Synthesis log; Q2iense measured cmpy_4_dsp48
Per-lane DSP cost 27–44	Measured 24. Estimate assumed plain radix-2.	Post-route utilisation
DSP and bandwidth both cap at ~4	DSP allows 9; bandwidth allows ~4. Bandwidth is 24 DSPs in series	24 DSPs in series
WHS +0.013 ns is not a pass	It is. Router fixes hold to just-positive by design.	report_timing -delay_type min

11 Limitations

- **Never run on physical hardware.** All results are simulation and Vivado static analysis.
- **Verification is narrow.** Structural checks plus peak-bin position on a single complex tone. No bin-by-bin golden-model comparison, no impulse, DC, two-tone, leakage, noise or low-amplitude cases.
- **Timing is out-of-context.** Clock skew is estimated rather than modelled on a real global buffer tree; hold margin must be re-confirmed after integration.
- **Only NUM_LANES = 1 has been built.** The generate loop is written but unexercised.
- **No window function.** A rectangular window has -13.3 dBc peak sidelobes, which will mask weak targets near strong ones. This belongs to the input stage, but the system needs it before any radar result is meaningful.
- **Complex multiplier structure is frozen at 4-multiplier.** This affects rounding as well as DSP count and must match project-wide, or tolerance mismatches appear later with no obvious cause.

12 Conclusion

The block meets its specification. It sustains 100 MSPS at one sample per clock, supports both transform sizes with live switching, forwards the block-floating-point exponent frame-aligned, handles backpressure correctly in both directions, and closes 100 MHz with 38 % margin while occupying about a tenth of the device.

The more useful conclusions are the negative ones. Custom FFT RTL is not worth building, because the vendor core is already radix-2² and would be matched, not beaten, after months of work including a block-floating-point reimplement. Further optimisation of this block is not worth doing either, because it is not the system bottleneck — interface bandwidth caps the design at roughly four lanes while the silicon would hold nine.

The work that remains is elsewhere: the window function, the corner-turn memory whose DDR bandwidth will decide the real system limit, and putting a design on the board.

All resource and timing figures in this report are measured from Vivado 2025.1 synthesis and implementation runs. All verification figures are from XSim behavioural simulation against the Xilinx FFT IP v9.1 simulation model. No number in this document is an estimate unless explicitly labelled as one.